Hologram — Resonant Homomorphic Framework — Math-Clean Revision

0. Overview

We formalize a compact algebra for resonance computation, a toggle-space model with a multiplicative evaluation map, a logic of resource budgets over $\mathbb{Z}_{96}$, and a homomorphic collapse to $\mathbb{Z}_2$. A factoring scaffold using homomorphic resonance is specified at the level of algebraic contracts and verifiable identities.

1. Algebraic base and notation

- Let $C := \mathbb{Z}_{96}$ with operations $\oplus$ (addition mod 96) and $\otimes$ (multiplication mod 96). $(C, \oplus, \otimes)$ is a commutative semiring with units 0 and 1 .
- Let $B := \mathbb{Z}_2^8$ be the 8-bit toggle space. Elements are written as bit-vectors $b = (b_0, \dots, b_7)$ or integers $b \in \{0, \dots, 255\}$. The group law is bitwise XOR, denoted $b \oplus_2 b'$.
- Parameters: positive reals $\alpha_i > 0$ with constraints
- $\alpha_0 = 1$,
- unity pair (α_4, α_5) with $\alpha_4 \alpha_5 = 1$.
- **Non-degeneracy.** Products formed from $\{\alpha_1, \alpha_2, \alpha_3, \alpha_6, \alpha_7\}$ are all distinct; moreover $\alpha_4 \notin \{1, \alpha_5\}$.

2. Resonance evaluation on toggles

Define the **pair-normalized evaluation** $R : B \to \mathbb{R}_{>0}$: for $b \in B$,

$$ R(b)\;:=\;\prod_{i\in{0,1,2,3,6,7}} \alpha_i^{b_i}\;\cdot\;\begin{cases} \alpha_4 & (b_4,b_5)=(1,0),\
1 & (b_4,b_5)\in{(0,0),(1,1)},\
\alpha_5 & (b_4,b_5)=(0,1).\end{cases} $$

This enforces the unity-pair constraint by using exponent $d(b) \in \{-1, 0, 1\}$ on α_4 with $\alpha_5 = \alpha_4^{-1}$. $\alpha_0 = 1$ makes bit 0 neutral.

Homomorphism on restricted groups. If $H \le B$ satisfies $b_4 = b_5$ for all $b \in H$, then $R(b \oplus_2 b') = R(b)R(b')$ for $b, b' \in H$.

3. Image size and basic statistics

Theorem (R96). Under non-degeneracy, $|R(B)| = 96$.

Reason. Bits $\{1, 2, 3, 6, 7\}$ contribute 2^5 distinct products. The pair (b_4, b_5) yields three distinct multiplicative states $\{\alpha_4, 1, \alpha_5\}$. Total $3 \cdot 2^5 = 96$.

Fairness schedule. One cycle enumerates $b=0,1,\dots,255$. Three cycles give 768 steps. Define

$$ \mu:=\frac{1}{256}\prod_{i=0}^7(1+\alpha_i),] where the product uses $\alpha_5=\alpha_4^{-1}$ and $\alpha_0=1$. The per-cycle variance is

$$\mathrm{Var}=\frac{1}{256}\prod_{i=0}^{7}(1+\alpha_i^2)-\mu^2.$$

4. Homomorphic substructures in B

Let $S:=\{0,1,2,3,6,7\}$. For any subgroup $T\le\mathbb{Z}_2^S$, the set [H_T:={b\in B: b_4=b_5\text{ and }(b_i)_{i\in S} \in T} $$

is a subgroup of B on which R is a group homomorphism into $(\mathbb{R}_{>0},\cdot)$. Examples include the trivial group, coordinate axes in S , and their direct sums.

5. Resonance Logic (RL)

Budgets. Associate to each sequent a budget $r\in C$. Primitive composition is $\mathrm{RI}:\quad r_1,\dots,r_n\ \vdash\ \rho:=r_1\otimes\cdots\otimes r_n\in C$. Structural rules use $\oplus$ for aggregation.

Klein complement. Define $\perp[r]:=[-r]_{96}$. Then $\perp(\perp[r])=r$ and $\perp[r\oplus s]=\perp[r]\oplus\perp[s]$. No general De Morgan law is claimed for $\otimes$.

Soundness contract. Any derivation computes its conclusion budget as the $\otimes$ -product of the premise budgets, up to commutativity and associativity in C .

6. Collapse semantics

Define the parity morphism $\kappa:C\to\mathbb{Z}_2$ by $\kappa([r]_{96})=r \bmod 2$. Then $\kappa(r\oplus s)=\kappa(r)\oplus_2\kappa(s),\qquad \kappa(r\otimes s)=\kappa(r)\cdot\kappa(s)$. Extend κ to sequents by replacing every numeric budget with its image.

Proposition (Homomorphic collapse). If a numeric RL derivation yields conclusion budget ρ from premises $r_1,\dots,r_n$, then the collapsed derivation over $\mathbb{Z}_2$ yields $\kappa(\rho)=\prod_k\kappa(r_k)$. Thus, parity-level proofs are images of numeric proofs.

7. Homomorphic Resonance Factorization (HRF-F)

This section specifies an abstract interface sufficient to design and test factoring pipelines without fixing an embedding.

7.1 Encodings and lifts

- **Encoding.** Choose a map $E : \mathbb{N} \to B^*$ into finite sequences of toggles. Requirement: $E(1) = \epsilon$ (empty sequence) and $E(mn) = E(m)\concat E(n)$.
- **Sequence evaluation.** Lift R to sequences by multiplicative accumulation:

$$ $$

- **Quantization to budgets.** Fix a quantizer $q : \mathbb{R}_{>0} \to C$. Define the budget of a sequence by $Q := q \circ R^*$.

7.2 Interference and accumulation

- **Histogram lift.** For a sequence $s \in B^*$, let $\mathrm{hist}(s) \in C^B$ record counts of each b reduced modulo 96.
- **Accumulator.** Choose weights $w \in C^B$ and define $\Phi(s) := \sum_{b \in B} w_b \otimes \mathrm{hist}(s)[b] \in C$.
- **Interference.** For sequences s, t , define $I(s,t) := \Phi(s) \oplus \Phi(t) \oplus \Phi(s\concat t)$. Other symmetric bilinear forms over C are admissible.

7.3 Factor signal

- **Design goal.** Select E, q, w so that for composite $N = pq$ with primes $p \neq q$, the statistic $I(E(p), E(q))$ separates from the null distribution induced by $E(r)$ for random r .
- **Verification protocol.** Specify a test function $\mathrm{match} : C \to \{0,1\}$ and evaluate false-positive/negative rates over sampled ranges. All thresholds are stated in C ; optional parity collapse via κ gives a binary surrogate.

8. Acceptance tests and reference identities

The following are implementation-level acceptance tests:

1. **R96 count.** With non-degenerate parameters, $|R(B)| = 96$; with any explicit degeneracy (e.g., $\alpha_1 = \alpha_2$) the count drops below 96.
2. **Triple-cycle identity.** Enumerated sum over three cycles equals $3 \prod_i (1 + \alpha_i)$; per-cycle mean and variance match the closed forms.
3. **Homomorphism checks.** For any subgroup H_T as in §4 and any $b, b' \in H_T$, $R(b \oplus_2 b') = R(b)R(b')$.
4. **Collapse homomorphism.** For random budgets $r, s \in C$, verify $\kappa(r \oplus s) = \kappa(r) \oplus_2 \kappa(s)$ and $\kappa(r \otimes s) = \kappa(r) \cdot \kappa(s)$; extend to RI compositions.

9. Appendices

A. Proof sketch for R96

Group the 256 bit-patterns by the pair (b_4, b_5) . Pair-normalization maps these four cases to three multiplicative states $\{\alpha_4, 1, \alpha_5\}$. The remaining five independent toggles generate 2^5 distinct products under non-degeneracy. Therefore the image size is $3 \cdot 2^5 = 96$.

B. Triple-cycle formulas

For one cycle, $\sum_{b \in B} R(b) = \prod_{i=0}^{7}(1+\alpha_i)$ by independence of bits under multiplication. The three-cycle constant is three times this value. The variance follows by replacing α_i with α_i^2 in the first moment and subtracting μ^2 .

If $b_4 = b_5$ and $b'_4 = b'_5$, then in pair-normalized evaluation the α_4/α_5 factors are both 1, and the map reduces to a pure product over indices in S . Because exponents add under XOR in $\mathbb{Z}_2$, multiplication of the corresponding α_i realizes the group law.

D. Collapse proposition

κ is a semiring morphism. Any numeric derivation in RL computes a polynomial in the premise budgets with coefficients in $\mathbb{N}$. Applying κ maps that polynomial to its image over $\mathbb{Z}_2$, yielding the stated result.

Implementation tip. Use pair-normalized evaluation for (α_4, α_5) to avoid numerical drift and to keep the three-state structure explicit.